



JDSU's MTS4000 FibreComplete optical test solution

error analysis, and bit- or symbol-error physical-layer analysis." In waveform measurements, various tests like accurate analysis of optical waveforms, basic eye-diagram and pulse waveform characterisation, jitter analysis and channel impedance characterisation are performed.

Optical testing equipment

Fibre-optical network testing involves a variety of equipment and applies to multiple functionalities and applications. Testing solution providers offer a range of equipment for the optical communications industry, for R&D, production, installation, monitoring and manufacturing.

Some of the common tools include OTDRs, BER sets, optical spectrum analysers (OSAs), power meters and light sources. Besides these, there are advanced-level tests, particularly in the areas of R&D and verification where complex and accurate test instruments are required.

Optical time-domain reflectometer. It is an optoelectronic instrument used

to characterise an optical fibre. An OTDR injects a series of optical pulses into the fibre under test. It may be used for estimating the fibre's length and overall attenuation, including splice and mated-connector losses. It may also be used to locate faults, such as breaks, and measure optical return loss.

Test and measurement (T&M) vendors provide a range of OTDRs in hand-held, bench-top and OEM module configurations. Some companies provide specialised OTDRs known as 'coherent OTDR.' These are used for locating faults in under-sea optical cables of up to 10,000 km between two countries or continents.

Optical spectrum analyser. Also known as spectrometer, it is used to measure the properties of light over a specific portion of the electromagnetic spectrum. It also allows characterisation of sources and DWDM systems for their power, bandwidth and performance.

Optical power meter. T&M vendors provide a range of optical power

meters including single-channel and multi-channel solutions for both hand-held and benchtop applications.

Optical component analyser. It is used to measure CD, PMD and PDL of the optical fibre and components. The lightwave component analyser is the instrument used to test all the relevant opto-electronic S-parameters and electro-optical components in 10 Gbps and 40 Gbps, fibre channel and CATV transmission systems as well as radio-over-fibre.

Bit-error rate test (BERT) sets. While dealing with any digital communications system, measurement of BER is very critical. Highly accurate and reliable BERT sets are used in high-speed, high-bandwidth optical communications.

Optical loss test sets. Optical test sets are multi-purpose optical measuring instruments that provide superior accuracy and reliability supporting reference light sources.

Optical signal sources/laser sources. Optical light sources are used to evaluate a wide range of optical devices and systems including WDM.

Multi-layer network test platform. It is designed to maximise the network performance and accelerate deployment of new services. Critical network parameters and performance can be monitored and network impairments quickly identified to accelerate deployment of new services. Results can then be easily documented by the powerful Windows-based platform simplifying and reducing the non-revenue-generating activities associated with operating an optical network. A modular design with different-size bay adaptors allows you to customise your test set as needed, adding further value.

Optical test peripherals. There are many application-specific peripherals for optical network testing, including video inspection probes, optical fibre identifiers and visual fault locators.

Optical sampling modules in oscilloscopes. For transmitter and receiver testing of optical network element designs up to 100 Gbps, companies